

Unconstrained Optimization

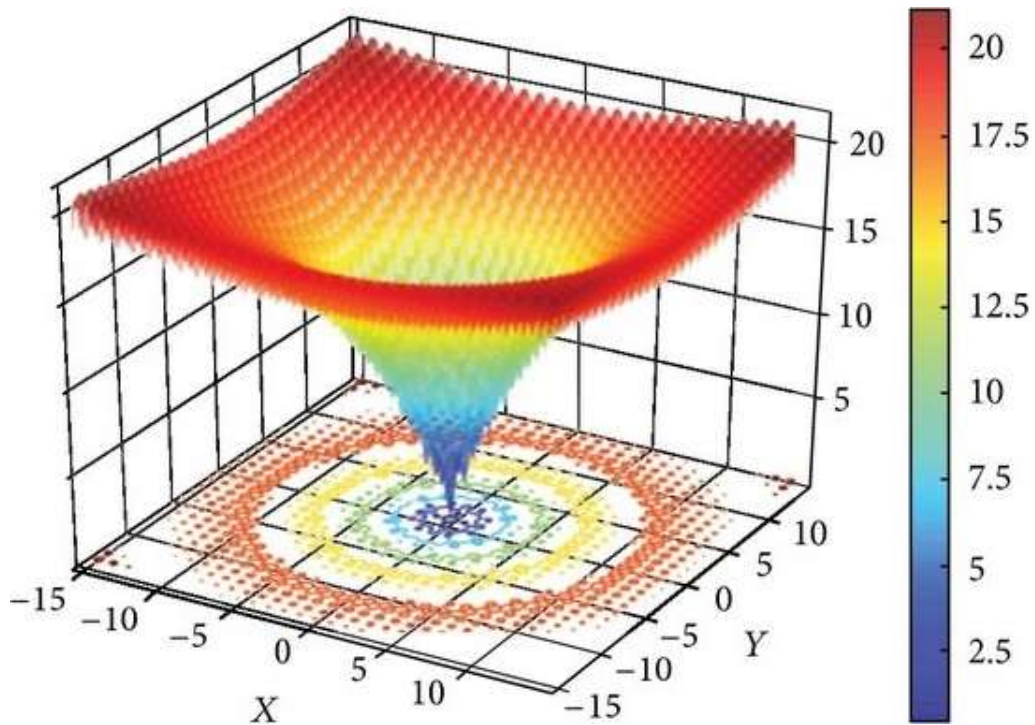


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Overview

- Introduction to Optimization
 - Applications & Evidences in Nature
 - Definitions & Formal Representations
 - Optimization Algorithms
- Single Variable Optimization: Bracketing Method
- Multiple Variable Optimization
- Gradient Descent & Steepest Descent
- Second Order Methods
- Solving Linear Regression?

Classical Optimization Problems



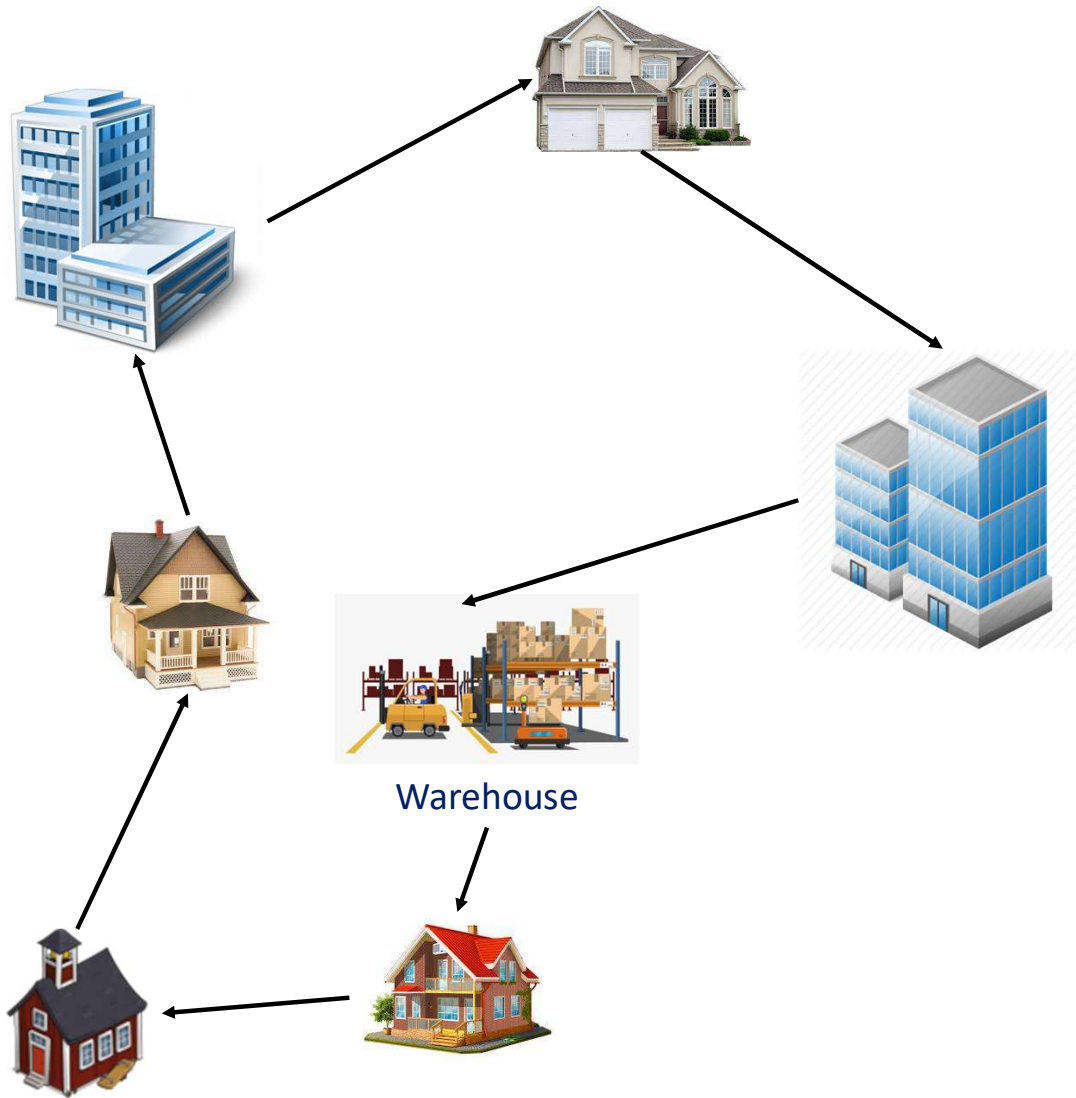
Minimize $z = f(x, y)$ w.r.t. (x, y)

$$z = -20e^{-g(x,y)} - e^{h(x,y)} + e + 20$$

$$g(x, y) = 0.2\sqrt{0.5(x^2 + y^2)}$$

$$h(x, y) = 0.5(\cos 2\pi x + \cos 2\pi y)$$

Global Optimum: $f(0,0) = 0$

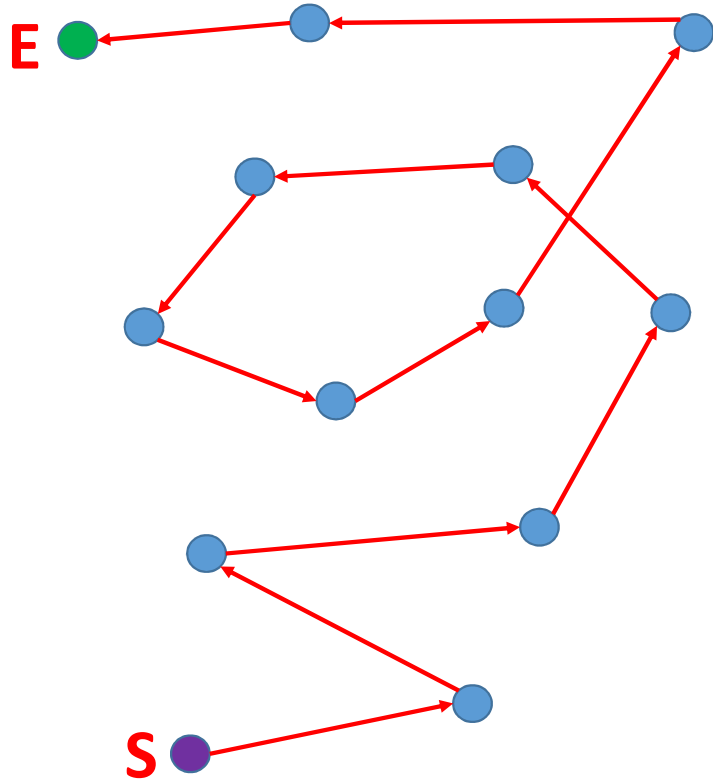


Visit All Delivery Locations

Minimize Travel Cost

Explore All Possible Paths

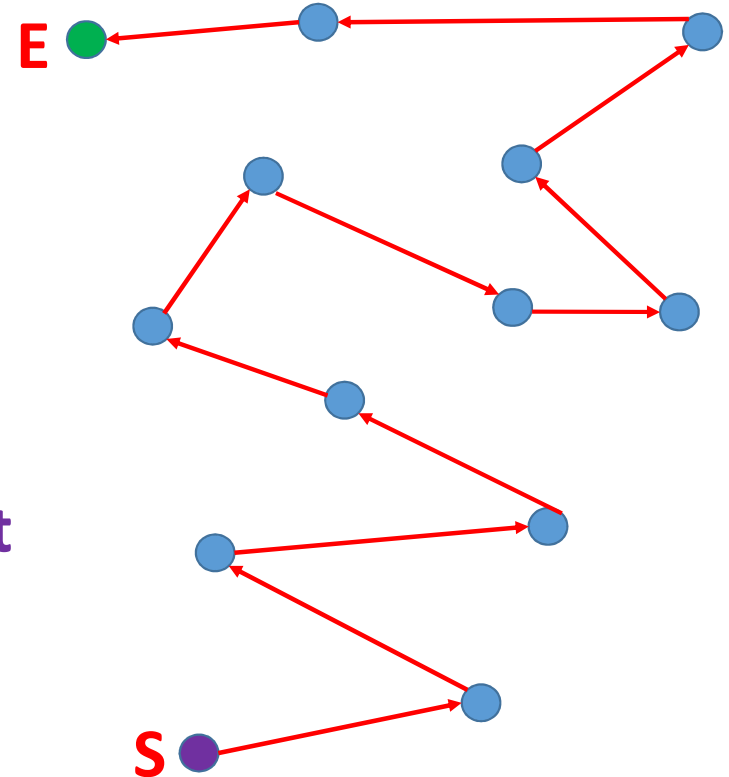
Travelling Salesman Problem



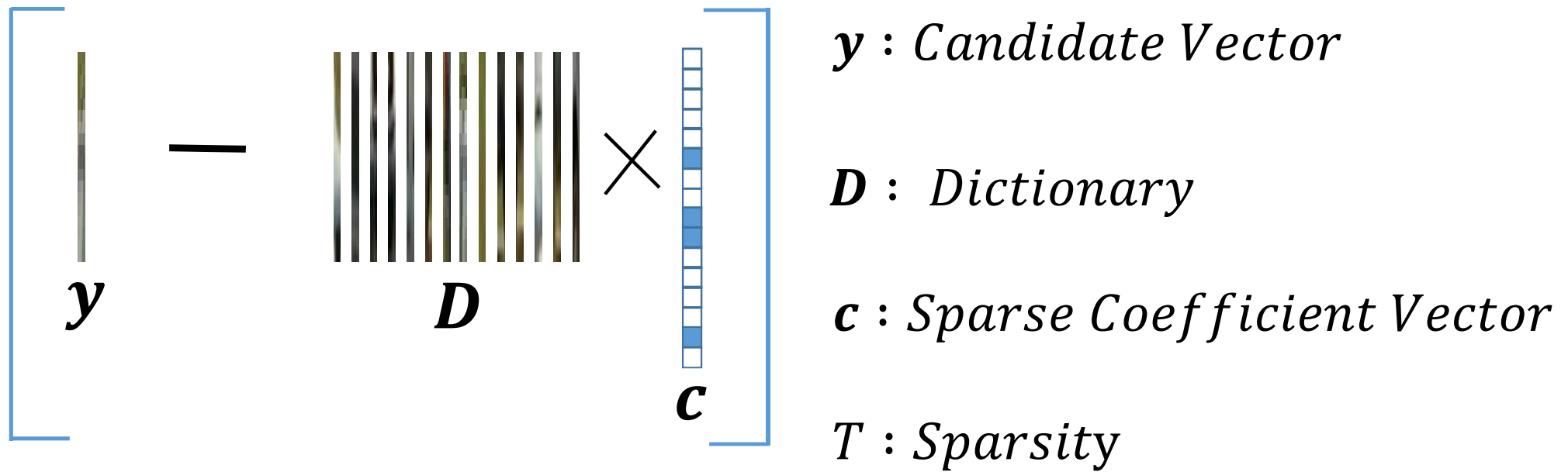
Visit All Stations
From **S** to **E**

Minimize Travel
Distance/Time/Cost

Explore
Station Sequences



Sparse Representation



$$\min_{\mathbf{c}} \|\mathbf{y} - \mathbf{D}\mathbf{c}\|_2^2 \text{ such that } \|\mathbf{c}\|_0 \leq T$$

Optimization: Definition & Formulation

Optimization is the Process of Maximizing or Minimizing one (or More) Desired Objective Function(s) while Satisfying a Set of Prevailing Constraints

Minimize $F_i(X); i = 1, \dots, m$

Subject To

$$g_p(X) \geq 0; p = 1, \dots, P$$

$$h_q(X) = 0; q = 1, \dots, Q$$

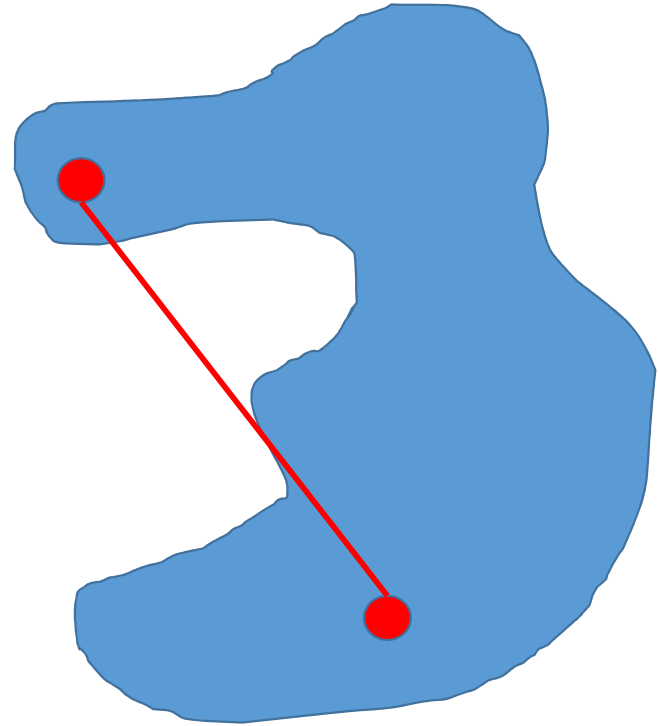
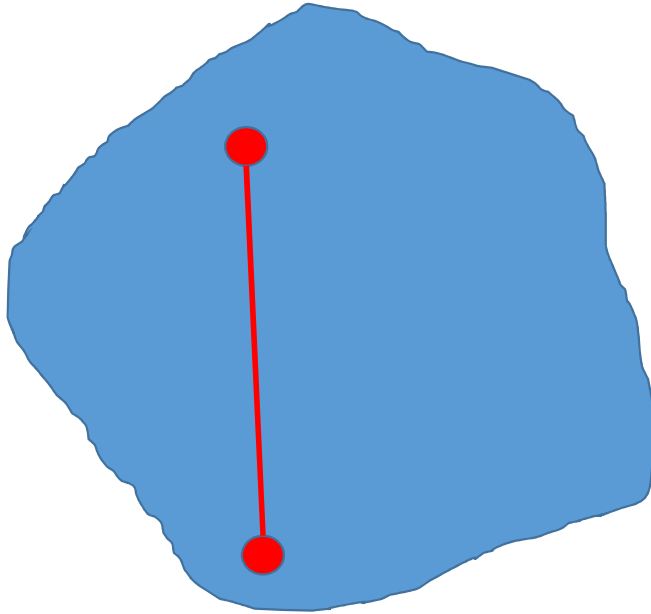
Optimization: Evidences in Nature

- Atoms assume positions of least energy to form unit cells in Metals & Alloys
- Liquid Droplet in Zero-Gravity is a Perfect Sphere
- Honeycomb as the Most Compact Packaging Structure
- Swarming Behavior of Insects – Ants & Fireflies (say)
- Genetic Crossover and Mutation Operations in Evolution

Optimization Algorithms

- Single Variable vs. Multiple Variable
- Unconstrained vs. Constrained Optimization
- Single Solution vs. Population based Approaches
- Classical Algorithms vs. Heuristic Approaches
- Nature Inspired Optimization Algorithms

Convex Set

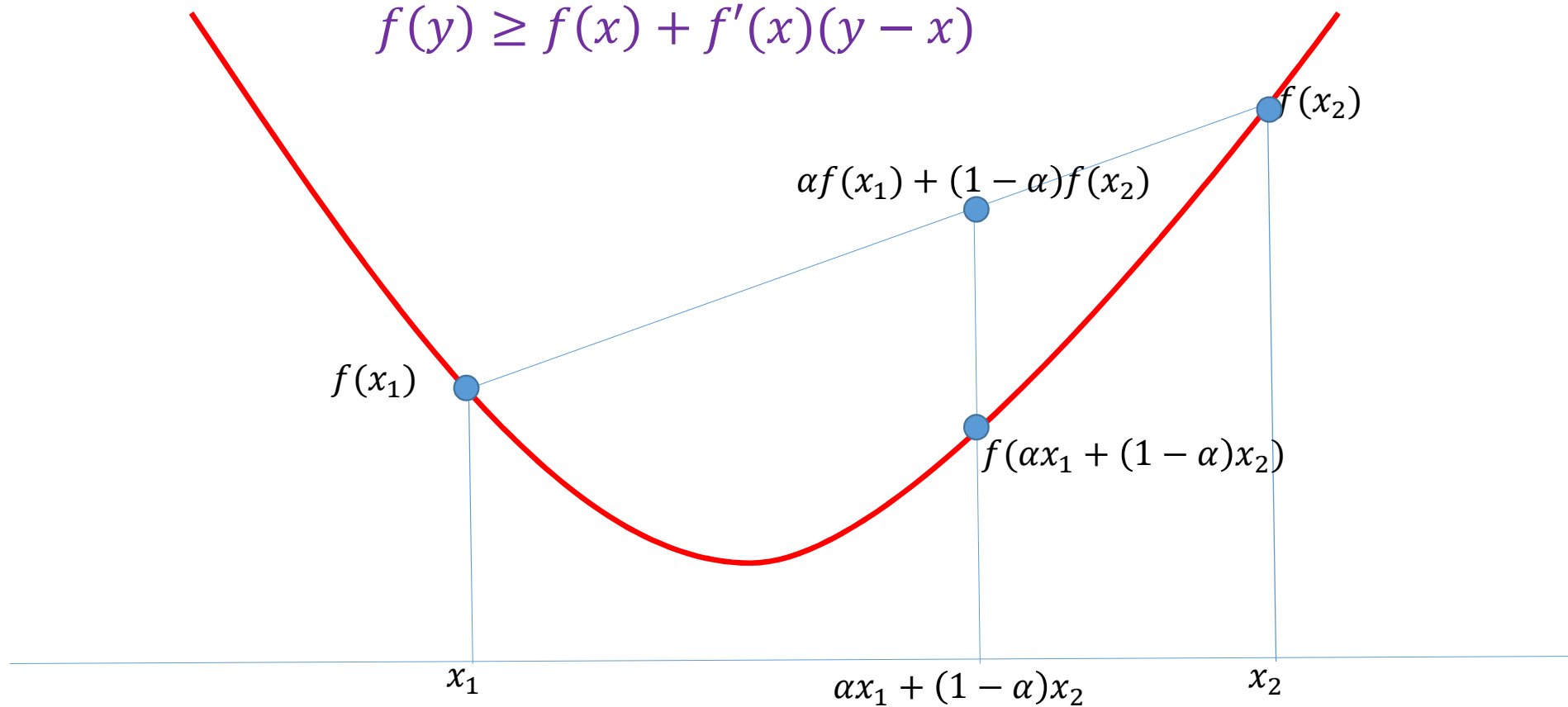


Convex Set: Line Joining Any Two Points Lie Within the Set

Convex Functions

$$\alpha f(x_1) + (1 - \alpha)f(x_2) \geq f(\alpha x_1 + (1 - \alpha)x_2)$$

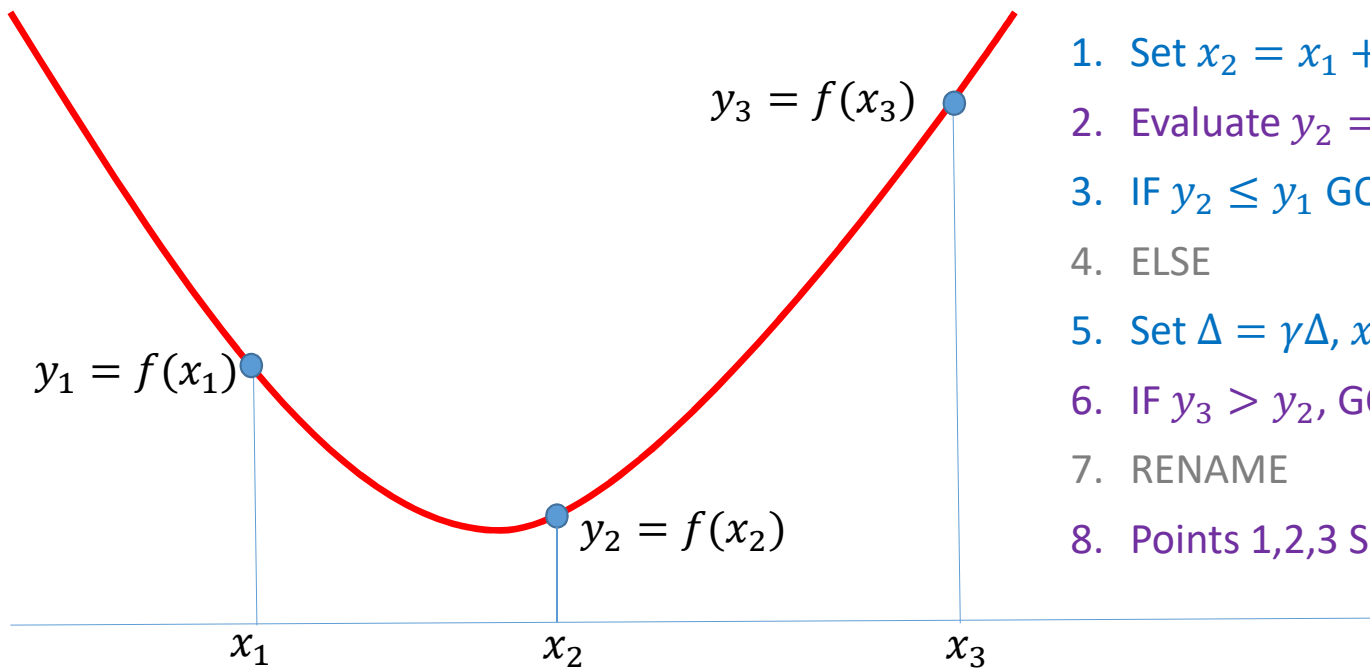
$$f(y) \geq f(x) + f'(x)(y - x)$$



SVO: Bracketing the Minima

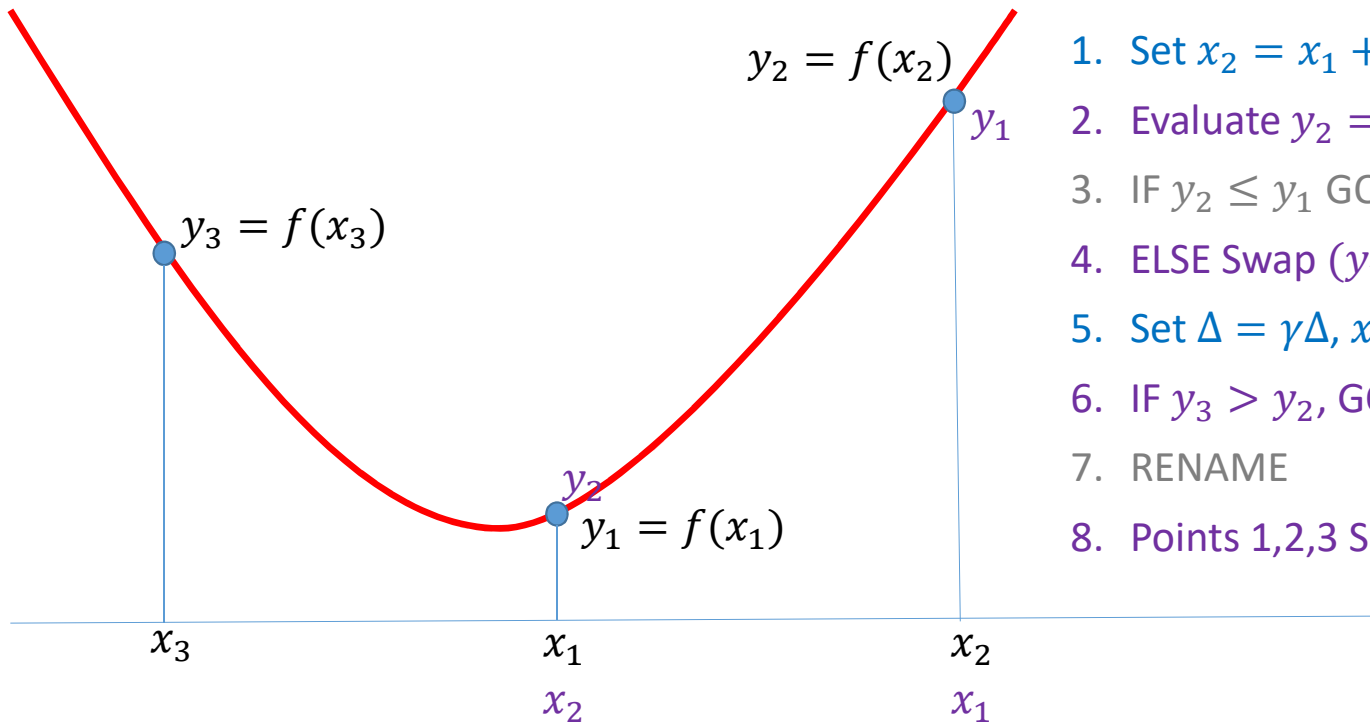
1. Set $x_2 = x_1 + \Delta$ (Step-Size)
2. Evaluate $y_2 = f(x_2)$ and $y_1 = f(x_1)$
3. IF $y_2 \leq y_1$ GOTO STEP 5
4. ELSE Swap (y_1, y_2) , (x_1, x_2) and Set $\Delta \leftarrow -\Delta$
5. Set $\Delta = \gamma\Delta$, $x_3 = x_2 + \Delta$ and evaluate $y_3 = f(x_3)$
6. IF $y_3 > y_2$, GOTO STEP 8
7. RENAME: $[y_2 \leftarrow y_1]; [y_3 \leftarrow y_2]; [x_2 \leftarrow x_1]; [x_3 \leftarrow x_2]$
AND GOTO STEP 5
8. Points 1,2,3 Satisfy $y_1 \geq y_2 < y_3$ Pattern

SVO: Bracketing the Minima



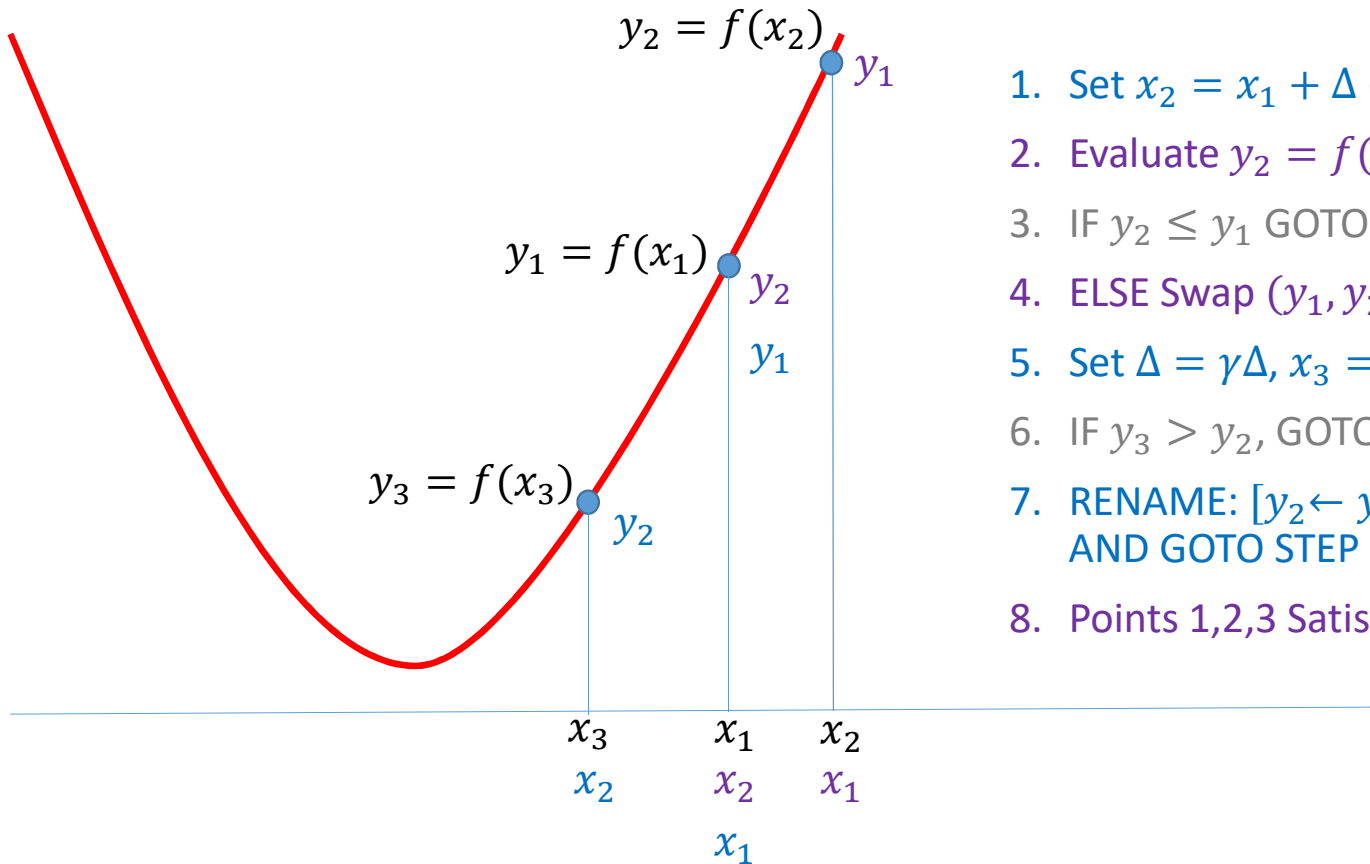
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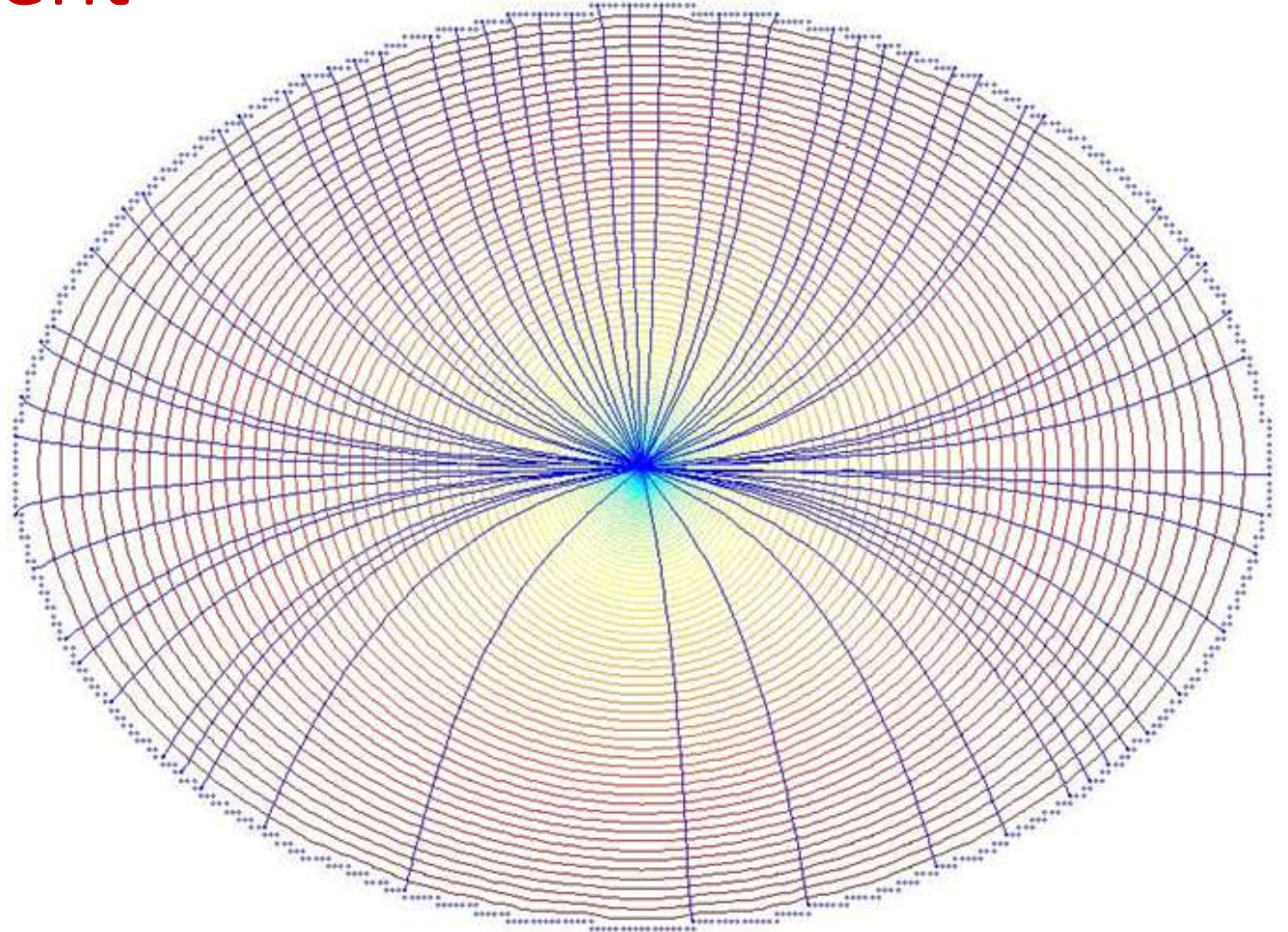
Gradient & Hessian

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

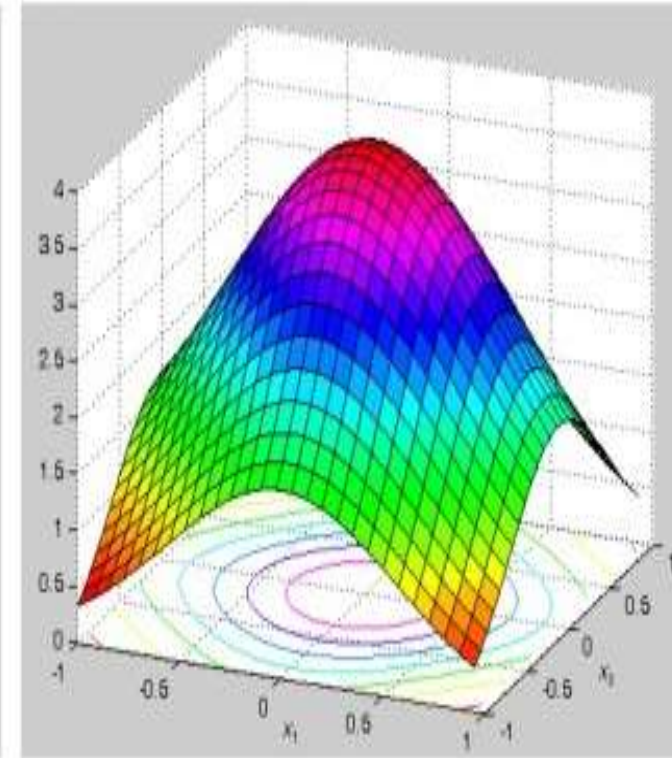
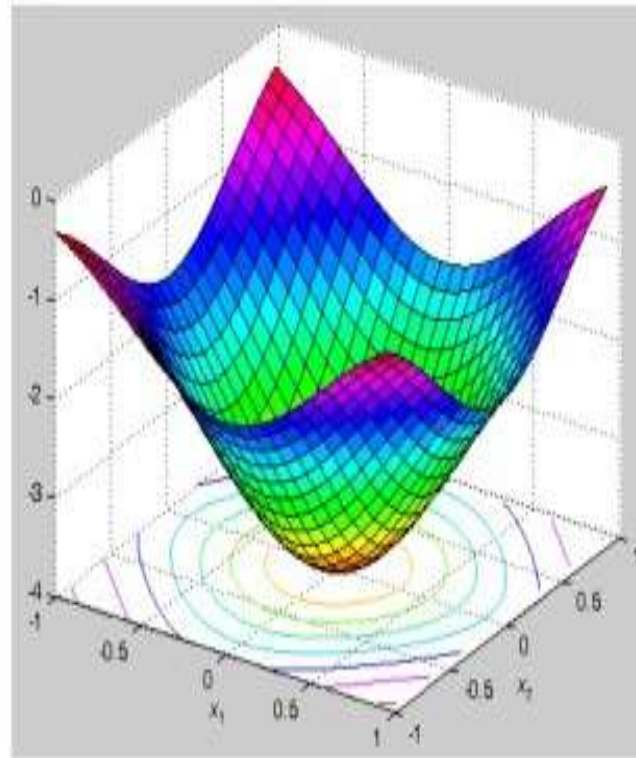
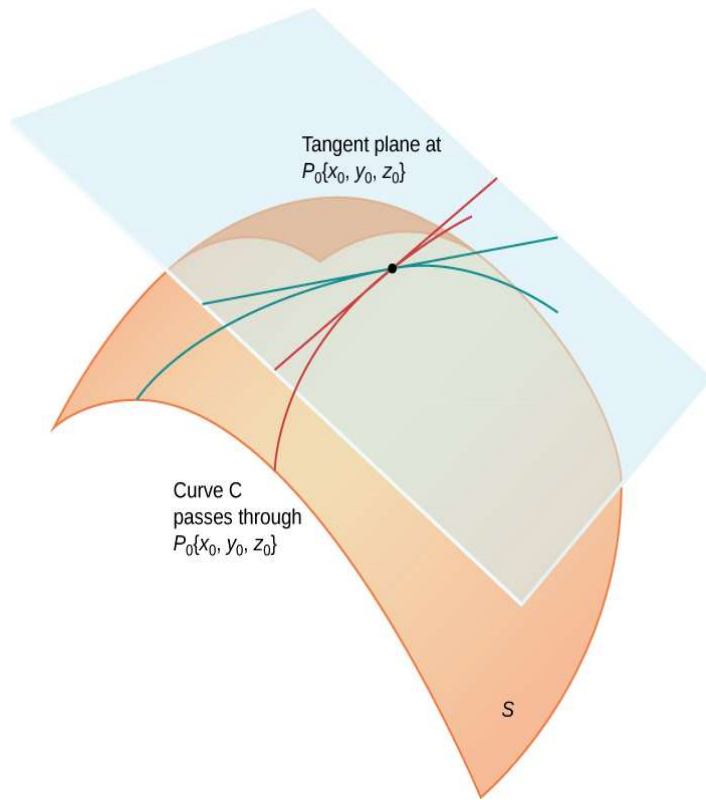
$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Revisiting Gradient

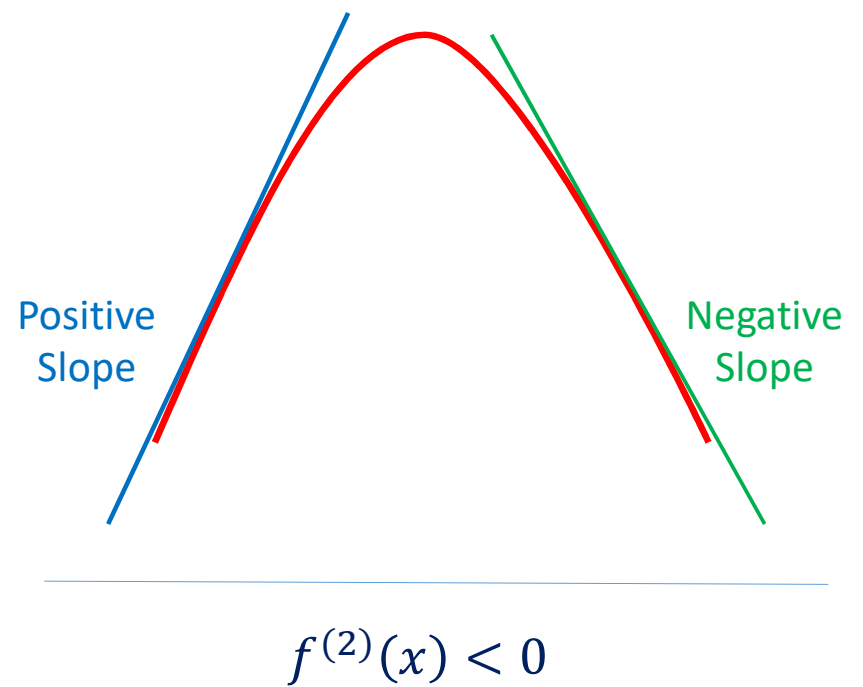
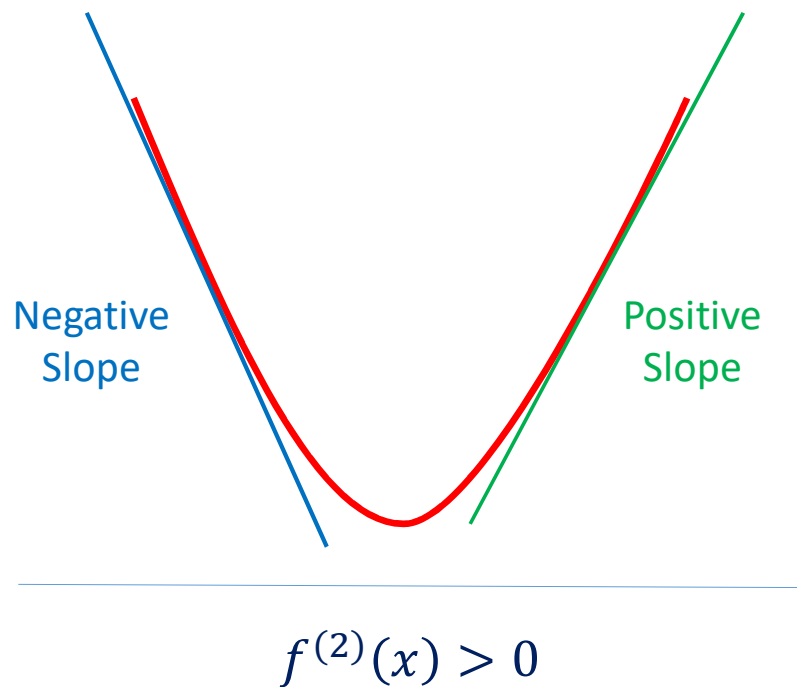
$$\nabla_x f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$



Gradient & Tangent Plane



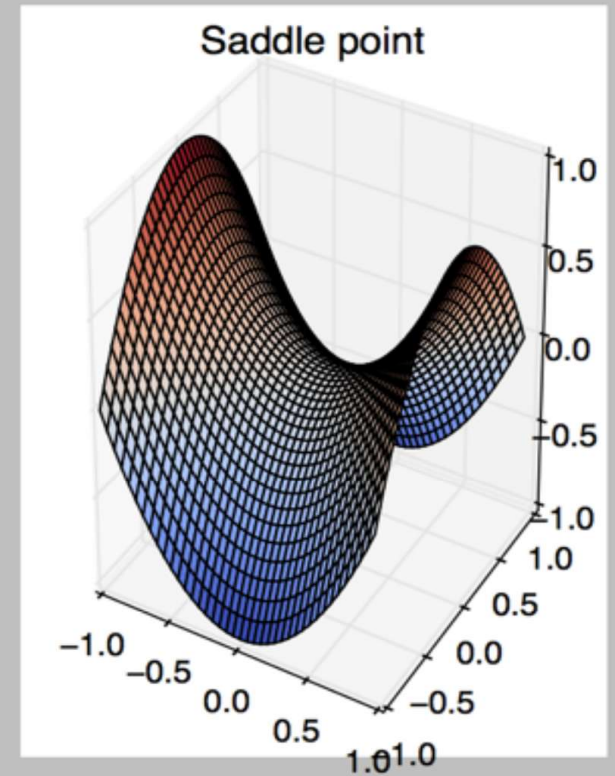
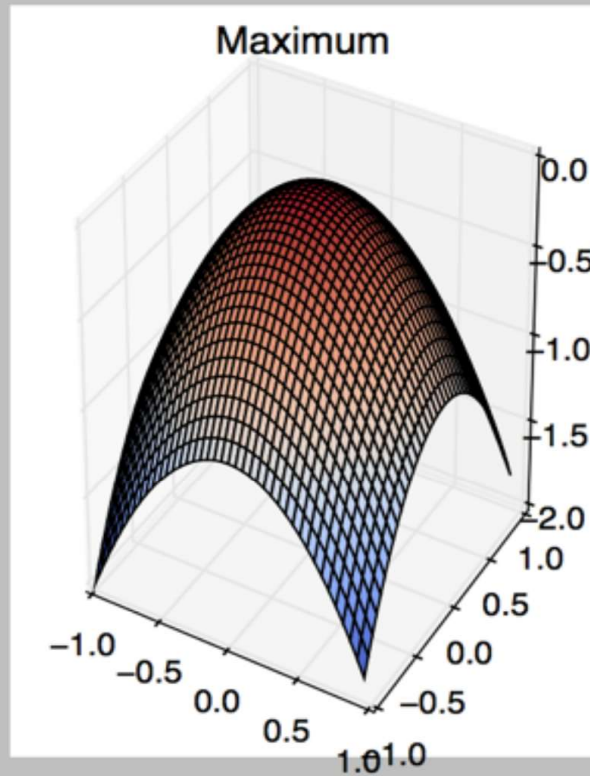
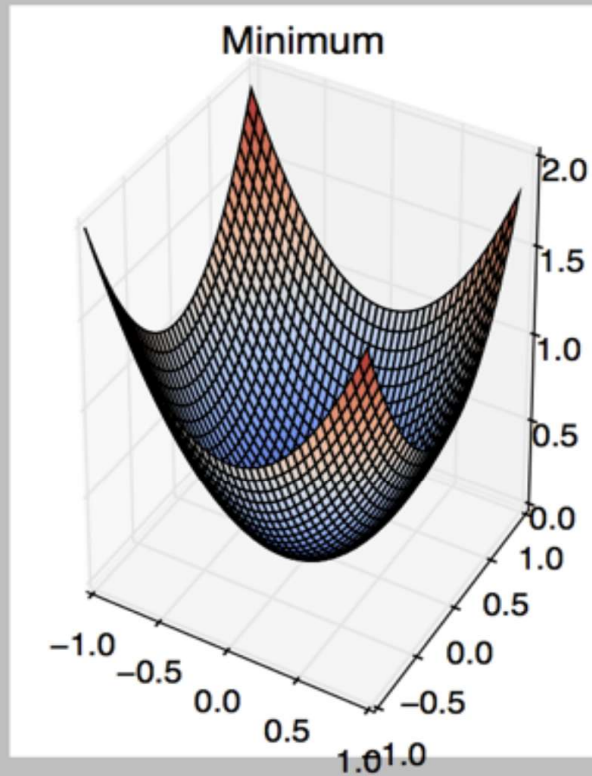
Role of $f^{(2)}(x)$ in SVO: $y = f(x)$



Hessian Revisited

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Significance of Hessian



All positive eigenvalues All negative eigenvalues

Some positive
and some negative

Optimality Conditions

Convex Functions

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \{\nabla f(\mathbf{x})\}^T (\mathbf{y} - \mathbf{x})$$

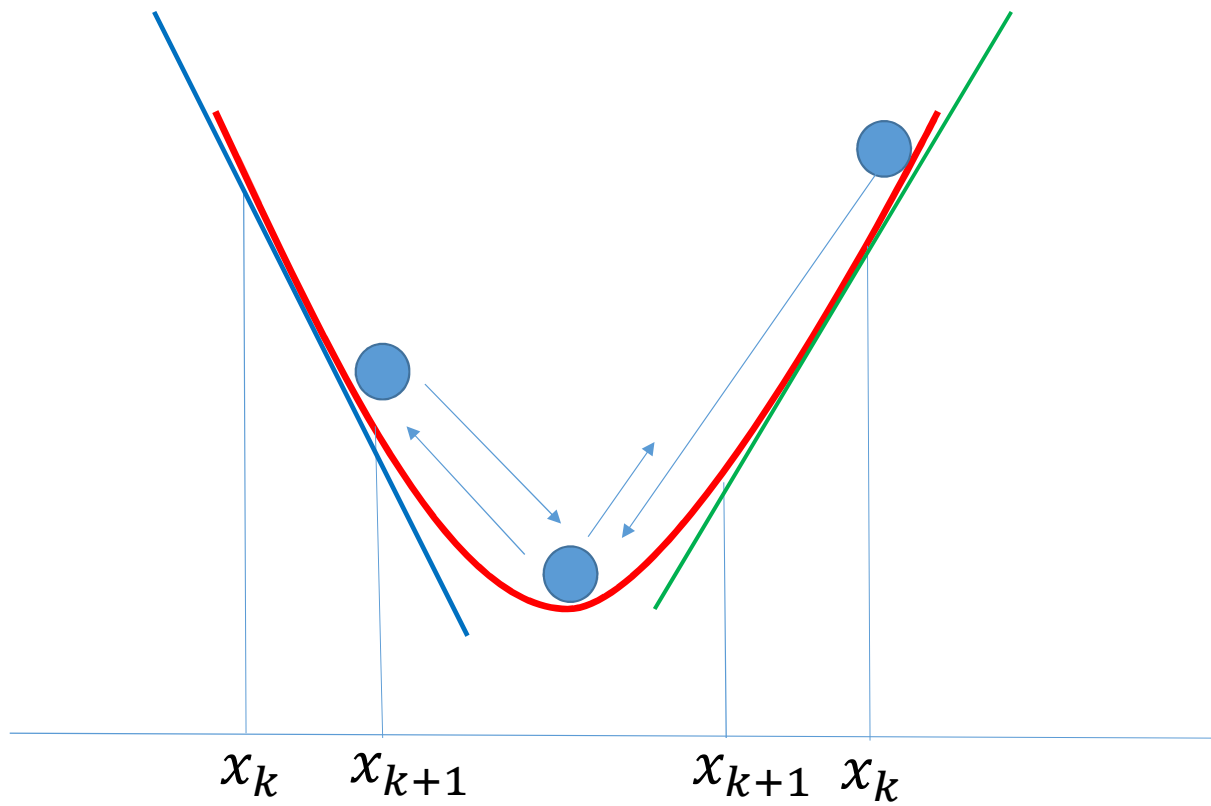
Necessary Conditions

$$\nabla f(\mathbf{x}) = \mathbf{0}$$

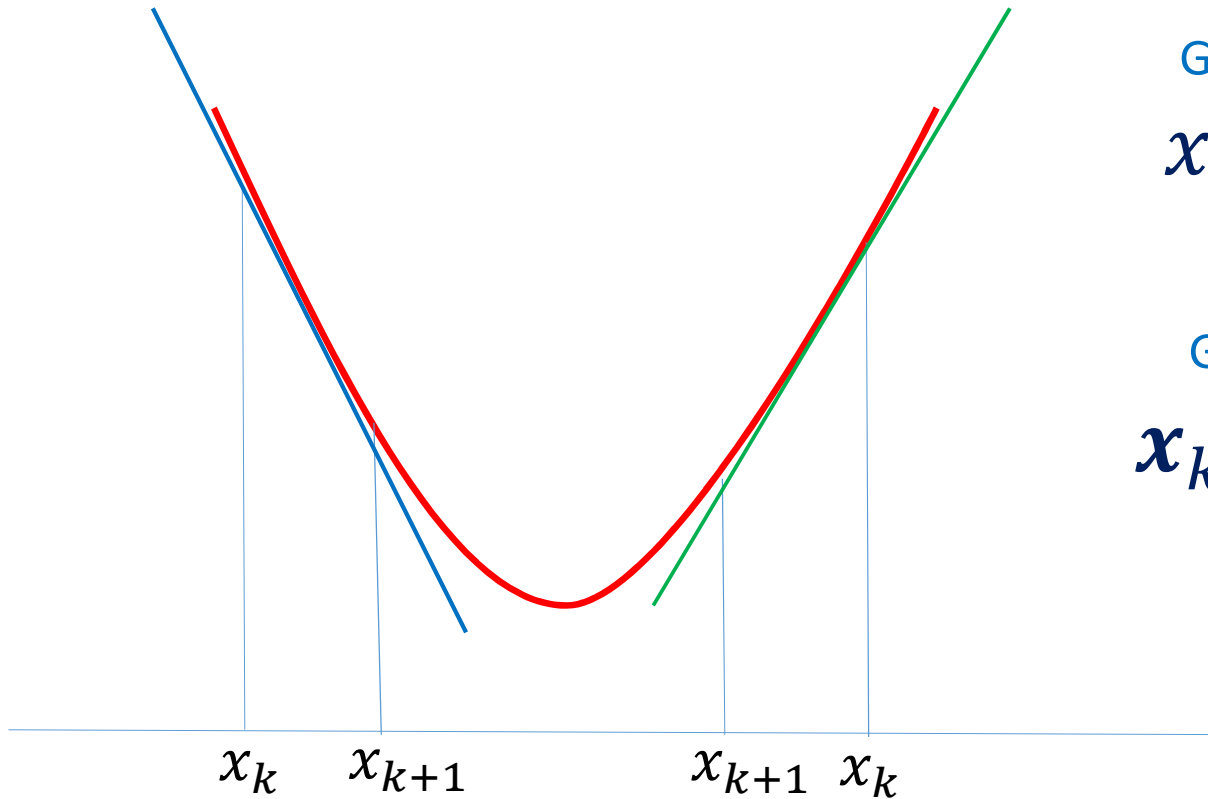
Sufficient Conditions

$$\mathbf{y}^T \left(\nabla^2 f(\mathbf{x}) \right) \mathbf{y} > \mathbf{0} \quad \|\mathbf{y}\|_2 \neq 0$$

Gradient Descent



Gradient Descent



Gradient Descent: Single Variable

$$x_{k+1} = x_k - \eta f'(x_k)$$

Gradient Descent: Multi-variable

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla_{\mathbf{x}} f(\mathbf{x}_k)$$

Oscillations: $\eta \leftarrow \frac{\eta}{2}$

Gradient Descent: Problem

Consider the following function

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2$$

Initial Value: $\mathbf{x}^{(0)} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \eta = 0.01$

Evaluate: $f(\mathbf{x}^{(0)}), \nabla_{\mathbf{x}}f(\mathbf{x}^{(0)}) \Rightarrow \mathbf{x}^{(1)} \Rightarrow \nabla_{\mathbf{x}}f(\mathbf{x}^{(1)}), f(\mathbf{x}^{(1)})$

Gradient Descent: Problem

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2 \quad \longrightarrow \quad \nabla f(\mathbf{x}) = \begin{bmatrix} 1 + 2x_2 + 4x_1 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

$$\text{Initial Value: } \mathbf{x}^{(0)} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \eta = 0.01$$

$$f(\mathbf{x}^{(0)}) = 33$$

$$\nabla f(\mathbf{x}^{(0)}) = \begin{bmatrix} 15 \\ 11 \end{bmatrix}$$

$$\mathbf{x}^{(1)} = \begin{bmatrix} 0.85 \\ 4.89 \end{bmatrix}$$

Gradient Descent: Problem

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2 \quad \longrightarrow \quad \nabla f(\mathbf{x}) = \begin{bmatrix} 1 + 2x_2 + 4x_1 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

$$\text{Present Value: } \mathbf{x}^{(1)} = \begin{bmatrix} 0.85 \\ 4.89 \end{bmatrix}, \eta = 0.01$$

$$f(\mathbf{x}^{(1)}) = 29.63$$

$$\mathbf{x}^{(2)} = \begin{bmatrix} 0.7082 \\ 4.7852 \end{bmatrix}$$

$$\nabla f(\mathbf{x}^{(1)}) = \begin{bmatrix} 14.18 \\ 10.48 \end{bmatrix}$$

$$f(\mathbf{x}^{(2)}) = 26.602$$

Steepest Descent

Gradient is Merely a Direction – We Do Not Know How Much to Walk in that Direction

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k) \quad \longrightarrow \quad \mathbf{d}_k = \frac{\nabla f(\mathbf{x}_k)}{\|\nabla f(\mathbf{x}_k)\|_2}$$

$$\text{Minimize SVO w.r.t. } \alpha \quad f(\mathbf{x}_k + \alpha \mathbf{d}_k) \quad \longrightarrow \quad \alpha^*$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha^* \mathbf{d}_k$$

Taylor Series

Single Variable: $y = f(x)$

$$f(a + h) = f(a) + hf^{(1)}(a) + \frac{1}{2}h^2f^{(2)}(a) + \dots$$

Multiple Variable: $y = g(\mathbf{x})$

$$g(\mathbf{a} + \mathbf{h}) = g(\mathbf{a}) + \mathbf{h}^T \{\nabla_{\mathbf{x}} g(\mathbf{a})\} + \frac{1}{2} \mathbf{h}^T \{\nabla_{\mathbf{x}}^2 g(\mathbf{a})\} \mathbf{h} + \dots$$

Second Order Methods: Quadratic Approximation

Quadratic Approximation around $\mathbf{x} = \mathbf{x}_k$

$$q(\mathbf{x}) = f(\mathbf{x}_k) + \{\nabla f(\mathbf{x}_k)\}^T (\mathbf{x} - \mathbf{x}_k) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_k)^T \{\nabla^2 f(\mathbf{x}_k)\} (\mathbf{x} - \mathbf{x}_k)$$



$$q(\mathbf{x}) \sim f(\mathbf{x}_k + \mathbf{x} - \mathbf{x}_k) = f(\mathbf{x}_k + \mathbf{h}) \text{ expanded up to two terms}$$

$$\mathbf{h}^* = \mathbf{x}_{k+1} - \mathbf{x}_k$$

Reference: Taylor's Series Expansion for Multiple Variables up to Two Terms

$$f(\mathbf{a} + \mathbf{h}) = g(\mathbf{a}) + \mathbf{h}^T \{\nabla_{\mathbf{x}} f(\mathbf{a})\} + \frac{1}{2} \mathbf{h}^T \{\nabla_{\mathbf{x}}^2 f(\mathbf{a})\} \mathbf{h}$$

Second Order Methods: Newton's Direction

Quadratic Approximation around $\mathbf{x} = \mathbf{x}_k$

$$q(\mathbf{x}) = f(\mathbf{x}_k) + \{\nabla f(\mathbf{x}_k)\}^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T \{\nabla^2 f(\mathbf{x}_k)\} \mathbf{h}$$

Minimization of $q(\mathbf{x})$ w.r.t. the Change from $\mathbf{x}_k$

$$\nabla_{\mathbf{h}} q(\mathbf{x}) = \nabla f(\mathbf{x}_k) + \frac{1}{2} [\{\nabla^2 f(\mathbf{x}_k)\} + \{\nabla^2 f(\mathbf{x}_k)\}^T] \mathbf{h}$$

Second Order Methods: Newton's Direction

Minimization of $q(\mathbf{x})$ w.r.t. the Change from $\mathbf{x}_k$

$$\nabla_{\mathbf{h}} q(\mathbf{x}) = \nabla f(\mathbf{x}_k) + \{\nabla^2 f(\mathbf{x}_k)\}\mathbf{h}$$

Setting $\nabla_{\mathbf{h}} q(\mathbf{x}) = \mathbf{0}$

$$\mathbf{h}^* = -\{\nabla^2 f(\mathbf{x}_k)\}^{-1}\{\nabla f(\mathbf{x}_k)\}$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \{\nabla^2 f(\mathbf{x}_k)\}^{-1}\{\nabla f(\mathbf{x}_k)\}$$

Second Order Methods: Problem

Consider the following function

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2$$

Determine Newton's Direction at $\mathbf{x}^{(0)} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

Evaluate: $\nabla f(\mathbf{x}^{(0)}), \nabla^2 f(\mathbf{x}^{(0)})$

Second Order Methods: Problem

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2$$

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 1 + 2x_2 + 4x_1 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

Second Order Methods: Problem

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2 \longrightarrow \nabla^2 f(\mathbf{x}) = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 1 + 2x_2 + 4x_1 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

$$\mathbf{x}^{(0)} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

$$\nabla f(\mathbf{x}^{(0)}) = \begin{bmatrix} 15 \\ 11 \end{bmatrix}$$

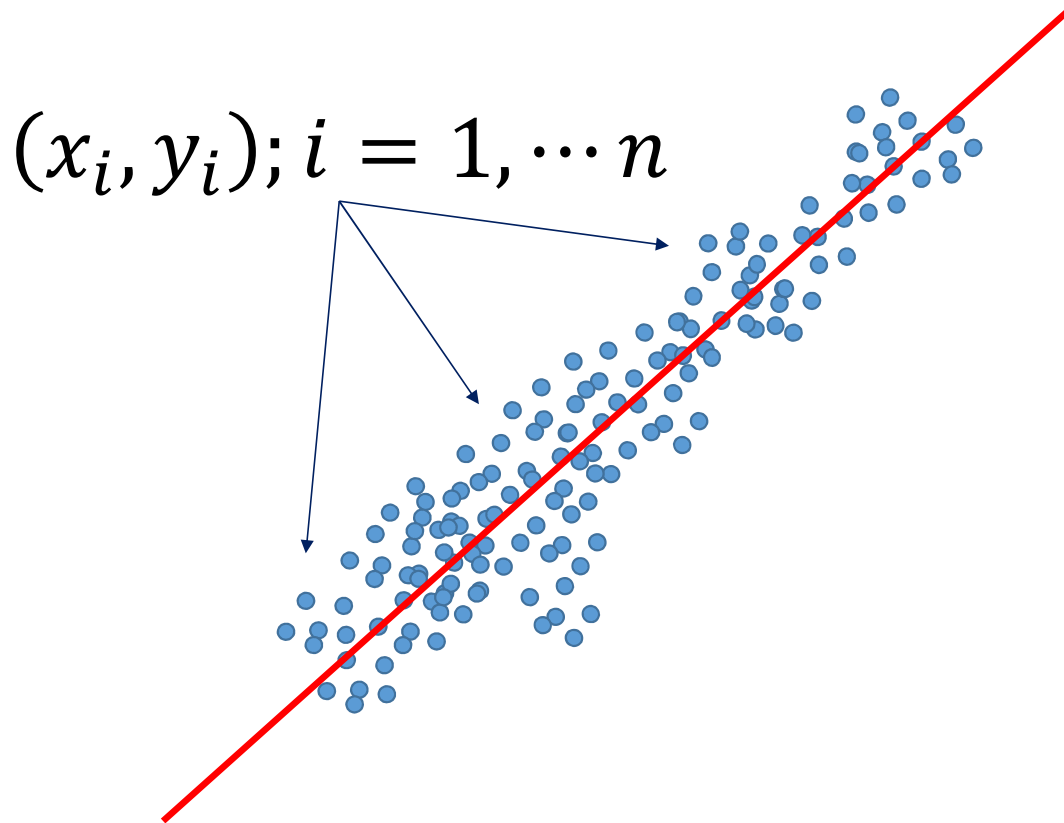
$$d_0 = -[\nabla^2 f(\mathbf{x}^{(0)})]^{-1}\{\nabla f(\mathbf{x}^{(0)})\}$$

$$d_0 = -\begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 15 \\ 11 \end{bmatrix}$$

$$d_0 = -\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1.0 \end{bmatrix} \begin{bmatrix} 15 \\ 11 \end{bmatrix}$$

$$d_0 = -\begin{bmatrix} 2.0 \\ 3.5 \end{bmatrix}$$

Linear Regression: Problem Formulation



Point Error

$$e_i = y_i - (a + bx_i)$$

Any value of (a,b) will provide us with some point error for given (x_i, y_i)

Total Error

$$E = \frac{1}{n} \sum_{i=1}^n e_i^2$$

Minimize E w.r.t (a,b)

Differentiating w.r.t. 'a' and 'b'

$$\frac{\partial E}{\partial a} = \frac{1}{n} \sum_{i=1}^n 2e_i \frac{\delta}{\delta a} \{y_i - a - bx_i\} = \frac{2}{n} \sum_{i=1}^n e_i \{-1\} = -\frac{2}{n} \sum_{i=1}^n \{y_i - a - bx_i\}$$

$$\frac{\partial E}{\partial b} = \frac{1}{n} \sum_{i=1}^n 2e_i \frac{\delta}{\delta b} \{y_i - a - bx_i\} = \frac{2}{n} \sum_{i=1}^n e_i \{-x_i\} = -\frac{2}{n} \sum_{i=1}^n \{y_i - a - bx_i\}x_i$$

$$a_{k+1} = a_k - \eta \frac{\partial E}{\partial a} (a = a_k)$$

$$b_{k+1} = b_k - \eta \frac{\partial E}{\partial b} (b = b_k)$$

Summary

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- Multiple Variable Optimization
- Gradient Descent & Steepest Descent
- Second Order Methods
- Linear Regression using Gradient Descent



Thank You